

HUMAN ACTIVITY RECOGNITION USING SMARTPHONE ACCELEROMETER DATA AND CNN-LSTM

Authors: NAVARRO, ROMEO V. | CAPISTRANO, IRISH L.

Year: 2024

Department of Computer Engineering

Degree Program: BSCpE

Keywords: human activity recognition, accelerometer, CNN-LSTM, wearable sensor, deep learning

ABSTRACT

This study proposes a CNN-LSTM hybrid model for human activity recognition using smartphone accelerometer data, achieving 97.1% accuracy on the UCI HAR dataset and 92.6% on a locally collected dataset.

INTRODUCTION

Human Activity Recognition (HAR) from wearable sensor data has applications in healthcare monitoring, elder care, and fitness tracking. Smartphones provide built-in accelerometers and gyroscopes, enabling unobtrusive monitoring without additional hardware. This study builds a CNN-LSTM model that combines CNN's spatial feature extraction with LSTM's temporal modeling for robust activity classification.

METHODOLOGY

The UCI HAR dataset (10,299 windows, 6 activities: Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, Laying) was used as the primary benchmark. A local dataset was collected from 20 CvSU students using their smartphones (Samsung and iPhone) in real-world conditions. Input: 128-timestep windows of 3-axis accelerometer and gyroscope signals (6 channels). Model: 2 Conv1D layers → MaxPool → 2 LSTM layers (128 units) → Dense. Training: 50 epochs, Adam, batch 64.

RESULTS

On UCI HAR: 97.1% accuracy. On the local dataset: 92.6% accuracy. The 4.5% performance gap is attributed to device heterogeneity (different sampling rates). Per-activity F1: Walking 99%, Sitting 96%, Laying 100%. Hardest class: Walking Downstairs vs. Walking (F1=94% and 95% respectively). Model size: 2.3MB, suitable for on-device inference.

DISCUSSION

The CNN-LSTM hybrid outperforms pure CNN (95.3%) and pure LSTM (96.1%) on UCI HAR by capturing both spatial signal patterns and temporal dynamics. Device heterogeneity is the most impactful real-world challenge. Future work should implement on-device inference using TensorFlow Lite, apply federated learning for privacy-preserving personalization, and expand the local dataset.